

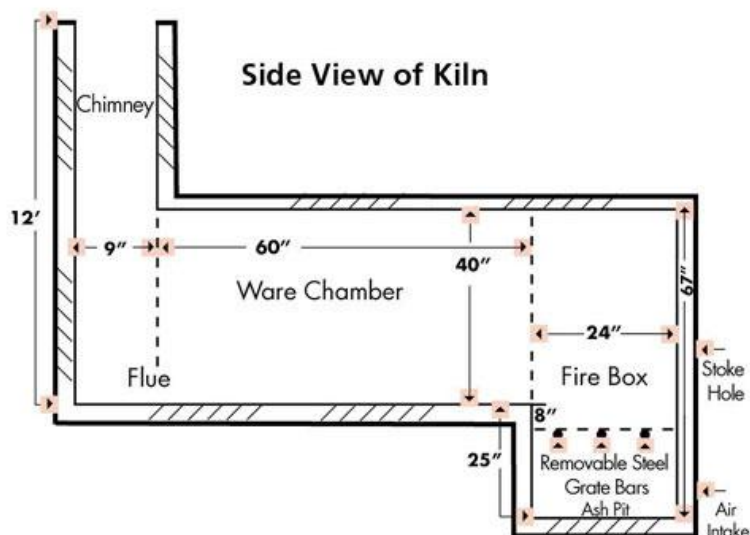
WOOD KILN PROJECT - 2024

(Or what happens when you decide to try the “fuck around and find out” method)

There are a few types of kilns. The earliest kilns were simply pits in the ground into which potters placed their pots. Early potters would cover the pots with combustible materials such as straws, dried manure and brush or sawdust, then light the whole pile and leave it to burn. Frederick Olsen, referring to a Japanese master potter, writes:

The late Tomimoto Kenkichi relates his theory on the development of the anagama (a Japanese word meaning "hole kiln"). Through the centuries, potters realized that the more enclosed the primitive kiln became, the hotter the firing and the more durable the pottery. Firing temperatures were increased by banking up the sides with clay Pottery villages were always built upon the clay source. (...) Utilizing the banks and hills of clay, they hollowed out holes large enough for a man to crawl into. The main cavity was widened and then tapered back into a chimney flue leading up to the surface. After the hole and cavity were hollowed out, the fire box, floor and walls were shaped in the clay and allowed to dry. A small fire of increasing intensity was built in the entrance or fire box and allowed to burn for weeks until the walls were bisqued. This transformed the chamber into a monolithic structure.

Anagamas are still used today in some oriental countries, along with other types of kiln, the noborigama (a multi chambered kiln), is still commonly used in Japan and elsewhere. Both of those styles, and any type of kiln that has a chimney on one side and the fire chamber on the other, are called *cross-draft* kilns. Cross-draft kilns allow the development of a special wood or ash effect, because the flame from the fire chamber, along with a lot of wood ash, are pulled through the pots in order to exit the chimney. This leaves ash on the pots, and when the built up gets thick enough. The ash melts and turns into glaze.

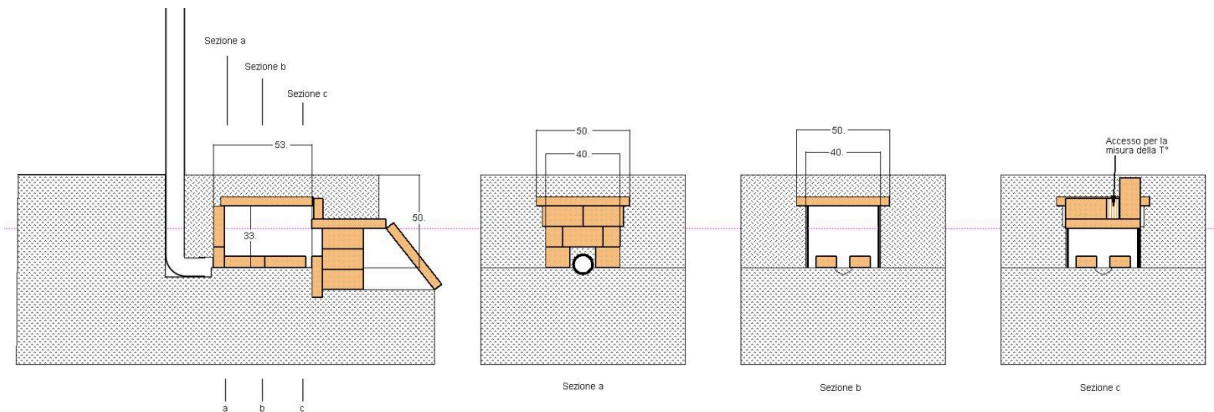


Another style of kiln is the *down-draft* kiln, which originated in Germany in the 1800's and it is one of the most commonly used kilns today. The heat source is located below or level with the base of the firing chamber. Flames come into the firing chamber and rise to the top of the kiln. The heat then rolls around the top and is sucked back down through the pots and out the flue. With the heat source under the firing chamber, most of the ash will stay in the fire box.

Manabigama: it is a cross-draft kiln built into the side of a hill. Its fire box is in the front, incorporated into the inside with a grate system, and extra air intakes are built into the front and sides. (Fires evenly to Cone 10-12 in eight hours tops).

Building the kiln

Based on the specifications of the Manabigama and the German downdraft kiln, we (my father and I) built a kiln set into the side of a hill. The kiln consists of a chimney, a raised firing chamber, and a firebox.



We started by digging into the side of the hill to create a vertical channel in which the kiln would be built. Once all the stones had been removed and the walls had been squared off as much as possible, the chimney flue was inserted. In this case, it was an old stove pipe approximately 230 cm long and 10 cm in diameter.



The refractory bricks were then positioned: four on the floor of the firing chamber, five at the back, around the chimney pipe, and another three separating the firing chamber from the firebox. The walls of the firing chamber were reinforced with two 3 mm-thick iron plates, which were then used to support a refractory clay slab approximately 5 cm thick.

The firebox was insulated with three bricks on each side, positioned horizontally, supporting a second refractory slab. Later, it became necessary to add two turf blocks to the two outer sides.

Once the structure was completed, we (I) fired the kiln for several hours in order to remove as much moisture as possible from the bricks and the earth.



Note to myself: next time you want to work on a project or build something, do it yourself. Whenever you ask for help, you end up doing whatever the other person wants, and the whole project derails into chaos.

The firing

On the evening of September 25th, I started firing the kiln. The fire was lit at 7:30 p.m. on the 25th and was continuously fed until 7:30 a.m. on September 26th.

It was regularly necessary to remove ash and embers from the firebox to make more space for the wood and allow more oxygen in (didn't think about that when I - uh, we - were building this thing). This was probably one of the factors that prevented the kiln from reaching the desired temperature (obviously).



Every hour the temperature was measured using a laser thermometer by removing a brick placed on top of the kiln (dumb decision) (red dot in the picture, if you can find it). This was DEFINITELY another factor that affected the temperature.

After 12 hours of firing (a nice barbeque and a cup of tea or ten) the kiln entrance was sealed with another refractory slab, on top of which a rock wood panel was then placed to

provide additional insulation. The top of the kiln was also covered with an additional layer of earth. (It looks fucking ugly, I know).



In total, the firing lasted almost three days. On the morning of September 28th, the temperature was low enough to remove the pieces without risking thermal shock and breakage. (Or so I thought, whoops).

NOTE - Things to improve:

Next time use a wider chimney and position it slightly higher.

Divide the firebox into two levels with a grate to make removing ash and charcoal easier and faster.

Improve the thermal insulation, and for the love of God, change the position of the brick used to check the temperature. Not on top of the kiln - the heat rises.

Use small, bone-dry wood (IMPORTANT)

Humid climate, average temperature: 19°C

Wood: wild cherry

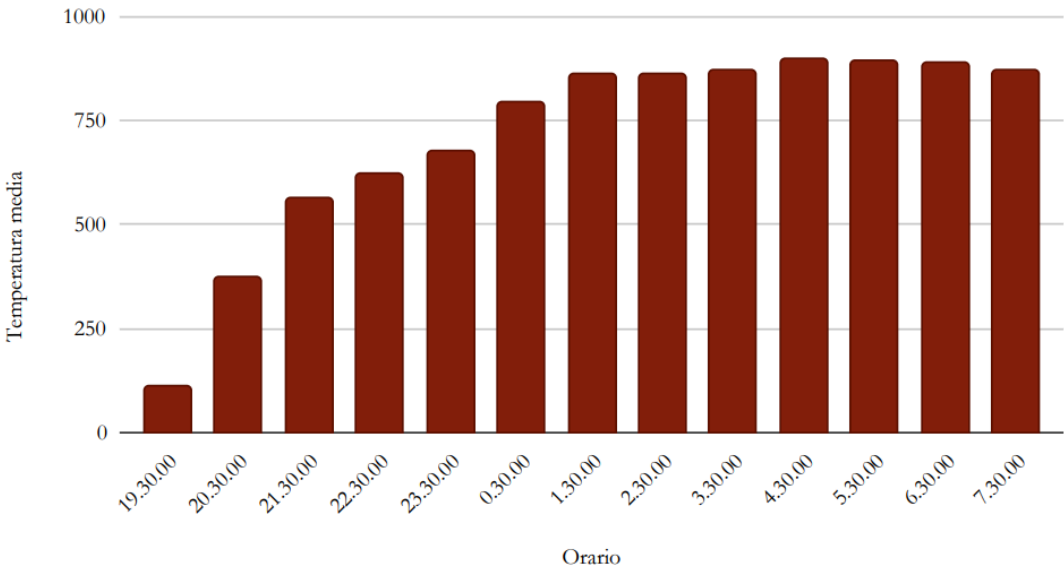
Firing started: 7:30 p.m., Wednesday, September 25

Kiln opened: 10 a.m., Saturday, September 28

Total firing: 62 h 30 min

Time	Temperature from above	Temperature from the front	Average temperature
19.30	112°C (circa)	-	112°C
20.30	336°C max 406°C	-	371°C
21.30	552°C max 574°C	-	563°C
22.30	607°C max 637°C	-	622°C
23.30	663°C max 691°C	-	677°C
00.30	779°C max 805°C	-	792°C
01.30	850°C max 871°C	-	860,5°C
02.30	850°C max 871°C	-	860,5°C
03.30	863°C max 878°C	(891°C max 921°C)	870,5°C
04.30	889°C max 909°C	(909°C max 913°C)	899°C
05.30	890°C max 898°C	-	894°C
06.30	887°C max 887°C	(906°C max 928°C)	887°C
07.30	865°C max 877°C	-	871°C

Temperatura media rispetto a orario



NOTES



The bronze glaze has the most severe defects: it detached from the ceramic surface and did not vitrify.

The turquoise glaze appears to be the most homogeneous. If the temperature had risen by another 80°C (approximately) it would probably have vitrified and produced a more conventional result.



The persian blue transparent glaze (above) formed lumps. In some areas, it also appears to have shrunk and cracked (it's called crawling, as you know, but I was a bit ignorant about this kind of thing at the time), creating a particularly interesting effect.

The clay collected from the river (top right) appears to be pinkish in color. It is excellent for throwing (on a banding wheel for now, I'll try it on a real throwing wheel when I have the opportunity), and fairly plastic (duh). It remains to be seen how it behaves with glazes and at high temperatures. (It melts at 1240°C).

